



# IndianOil

## JOURNEY RISK MANAGEMENT (JRM) STUDY

### Salem Terminal to SANKAGIRI LORRY OWENER'S ASSOCIAATI

#### Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

#### Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

#### Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

## Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

## Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

## Environmental Considerations

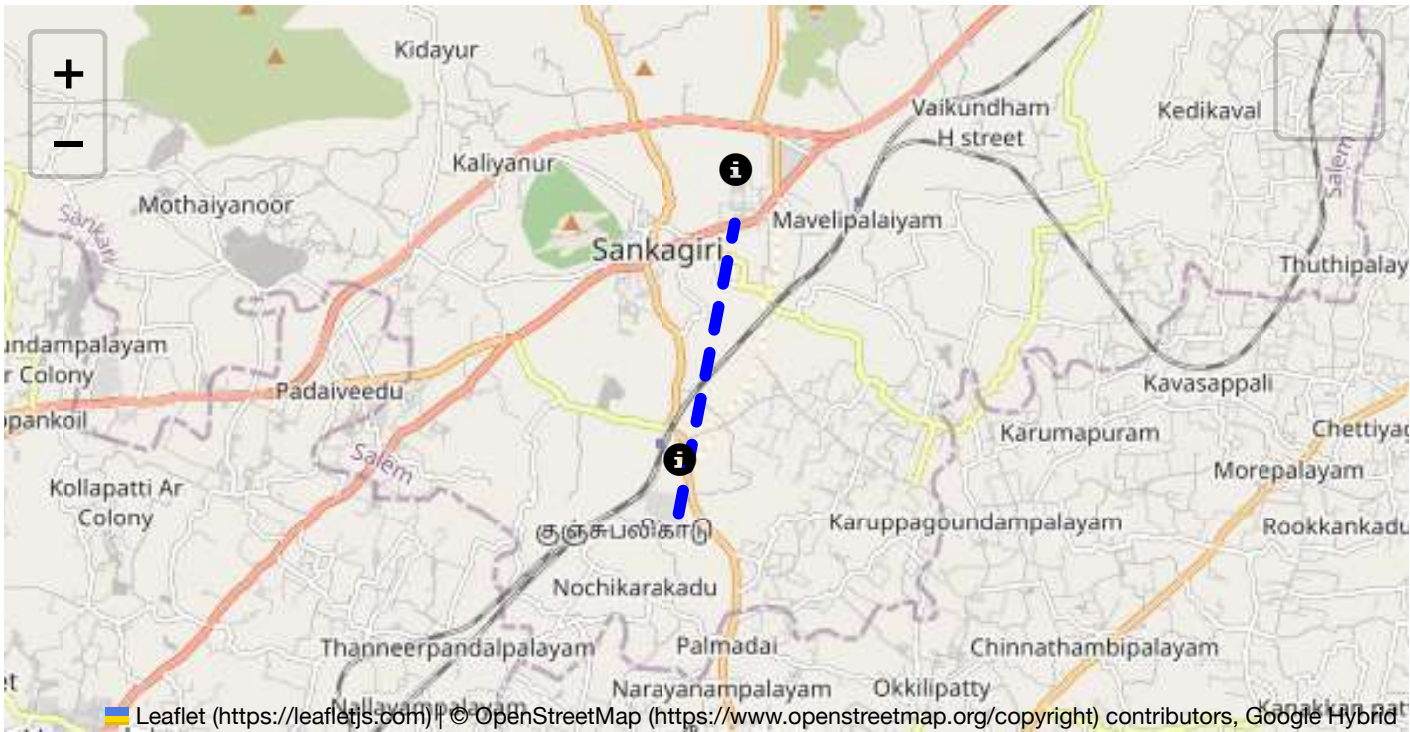
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

## Journey Risk Mitigation

The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

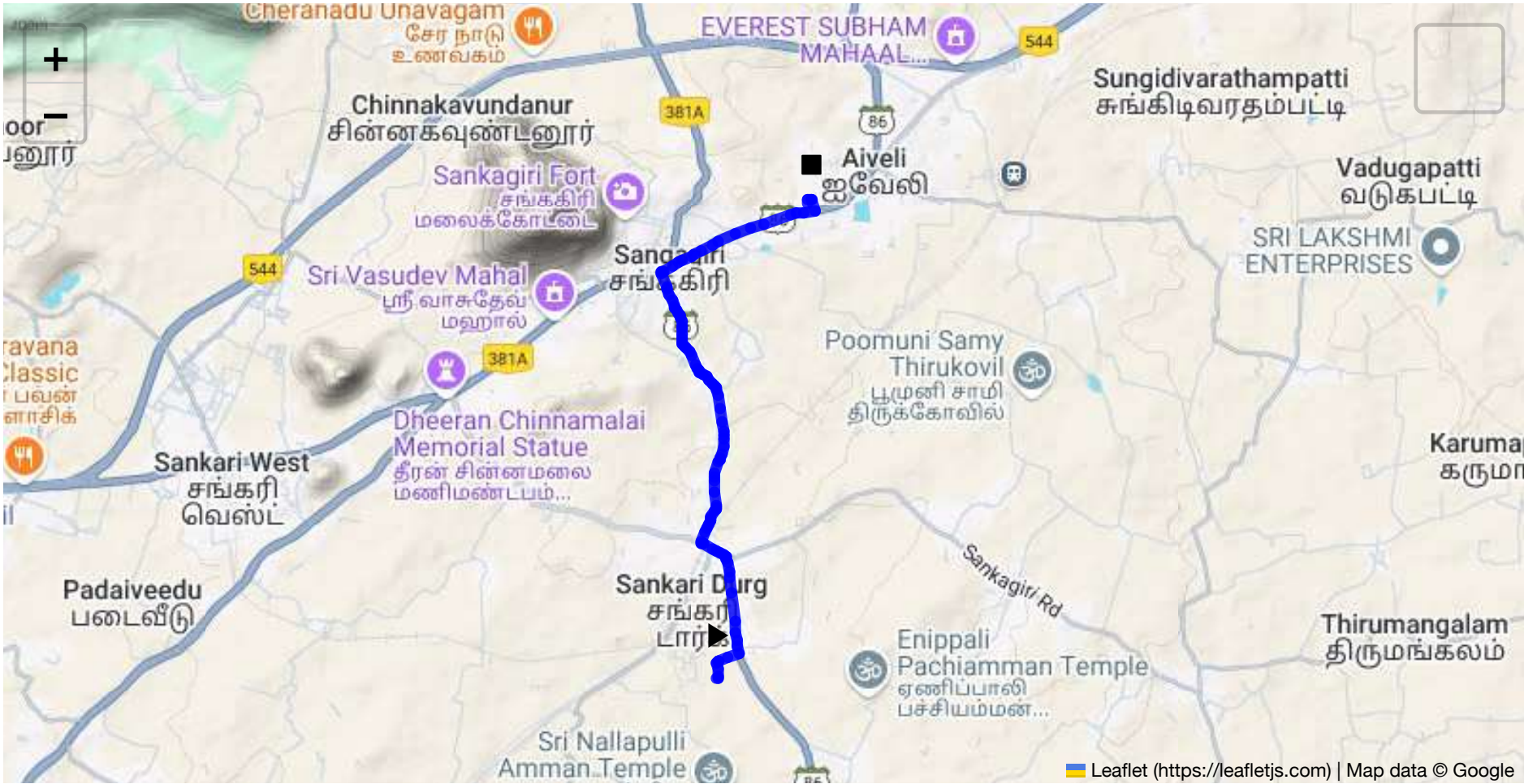
## Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



**Route Summary:**  
**Total Distance: 6.38 km**  
**Estimated Duration: 0.2 hours**

Adjusted Duration (Heavy Vehicle): 0.3 hours  
Start: (11.4381, 77.8734)  
End: (11.47982, 77.881706)



Welcome to the Journey Risk Management Study

Route Safety Analysis from CVQF+23W to FVHJ+XH6, Sangagiri, Tamil Nadu

- 1. Overview of the Route Map:
  - The route covers approximately 6.38 kilometers within Sangagiri, Tamil Nadu. It primarily involves local roads passing through both residential and semi-urban areas.
- 2. Typical Weather Conditions and Potential Weather-Related Hazards:
  - Sangagiri experiences a tropical climate with notable monsoon activity in June to September, leading to potential flooding and reduced visibility. During the summer, temperatures can rise significantly, causing heat distress and impacting road conditions.
- 3. Traffic Patterns:
  - Morning (8:00 AM - 10:00 AM) and evening (5:00 PM - 7:00 PM) are peak hours with increased congestion, particularly near market areas and schools.
  - Weekends may experience lighter traffic, though events or festivals could alter these patterns.
- 4. Assessment of Road Quality and Infrastructure:
  - Road conditions are generally good, but some stretches may have potholes and uneven surfaces. Infrastructure like signaling systems and signage are in place but should be checked regularly for functionality.
- 5. Suggestions for Alternative Routes for Emergencies:
  - Given the short distance, alternative routes may involve detours through smaller streets. However, these may not be suitable for heavy vehicles. An emergency plan should include verification of alternate paths or reliance on local authorities for guidance.

6. **Summary of Local Regulations Affecting Hazardous Material Transport:**
  - Strict timings are often enforced for the movement of heavy and hazardous material transport, typically avoiding peak school hours. Compliance with load limits and proper documentation is mandatory.
7. **Overview of Historical Incidents:**
  - There have been few incidents within the locality related to heavy vehicles, primarily due to overloading or mechanical failure. Attention to vehicle maintenance and adherence to load capacity is crucial.
8. **Environmental Considerations and Sensitive Areas:**
  - The route may pass through areas near schools and residential zones, where noise and pollution from hazardous material transports should be minimized. There are no major environmentally protected sites along the route, but precautions near water bodies should be maintained.
9. **Analysis of Communication Coverage:**
  - Generally good mobile network coverage throughout the route. However, passing through underpasses or specific spots with heavy foliage might temporarily affect signal quality.
10. **Estimated Emergency Response Times for Different Route Segments:**
  - Emergency services are estimated to reach most parts of the route within 15-20 minutes, subject to current traffic conditions. Proximity to local hospitals and fire services enhances rapid response ability.
11. **Overall Summary of Risk Assessment:**
  - The route presents moderate risk, primarily elevated during peak congestion or adverse weather conditions. Proactive vehicle maintenance, adherence to traffic laws, and strict compliance with hazardous material regulations will mitigate risks. Preparation for unexpected detours and maintaining effective communication channels are recommended. Overall, while the route is well-traveled and manageable, diligence ensures both safety and compliance.

Risk Assessment - Turns

|   | Risk Type  | Risk Level | Coordinates        | Speed Limit |
|---|------------|------------|--------------------|-------------|
| 0 | Turn       | High       | 11.43811, 77.87348 | 15 KM/Hr    |
| 1 | Turn       | Medium     | 11.43956, 77.87340 | 30 KM/Hr    |
| 2 | Turn       | Medium     | 11.43971, 77.87348 | 30 KM/Hr    |
| 3 | Turn       | High       | 11.44029, 77.87544 | 15 KM/Hr    |
| 4 | Turn       | High       | 11.45007, 77.87201 | 15 KM/Hr    |
| 5 | Turn       | High       | 11.47407, 77.86839 | 15 KM/Hr    |
| 6 | Blind Spot | Blind Spot | 11.47946, 77.88218 | 10 KM/Hr    |
| 7 | Turn       | High       | 11.48056, 77.88195 | 15 KM/Hr    |
| 8 | Turn       | High       | 11.48050, 77.88152 | 15 KM/Hr    |

Emergency Locations

|   | type     | name                    | coordinates            | speed_limit | risk_level |
|---|----------|-------------------------|------------------------|-------------|------------|
| 0 | police   | Police Quarters         | 11.4741243, 77.8673495 | 30 km/h     | Medium     |
| 1 | hospital | Kathir Eye Hospital     | 11.4740976, 77.8681591 | 30 km/h     | Medium     |
| 2 | hospital | Thilagam Hospital       | 11.4753573, 77.8685829 | 30 km/h     | Medium     |
| 3 | clinic   | Meyyazhagan Clinic      | 11.4752254, 77.8685671 | 30 km/h     | Medium     |
| 4 | hospital | Dr. Jaganathan Hospital | 11.4752717, 77.8709312 | 30 km/h     | Medium     |
| 6 | hospital | Sankari Hospital        | 11.4777496, 77.873892  | 30 km/h     | Medium     |
| 8 | clinic   | Dr. Dhanasekar Clinic   | 11.4793113, 77.8829247 | 30 km/h     | Medium     |
| 9 | hospital | Sri Vellaiyan Hospital  | 11.4798876, 77.8831394 | 30 km/h     | Medium     |

Crowded Spots

|   | type        | name                                   | coordinates            | speed_limit | risk_level |
|---|-------------|----------------------------------------|------------------------|-------------|------------|
| 5 | school      | Goverment Boys Higher Secondary School | 11.477659, 77.8659632  | 30 km/h     | Medium     |
| 7 | marketplace | Sunday Market                          | 11.4796502, 77.8708458 | 30 km/h     | Medium     |

Route Photos of Risky Spots



Risk Type: Turn  
Risk Level: High  
Speed Limit: 15 KM/Hr  
Coordinates: 11.44029, 77.87544



**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 11.45007, 77.87201



**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 11.47407, 77.86839



**Risk Type:** Blind Spot  
**Risk Level:** Blind Spot  
**Speed Limit:** 10 KM/Hr  
**Coordinates:** 11.47946, 77.88218

